Alzheimer’s Dementia

Kimberly Eyer, Karen Pendergrass, Dr. Umar

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Abstract. Explore Alzheimer’s disease and how gut dysbiosis and microbial metabolites may drive neuroinflammation and cognitive decline.

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Overview

Alzheimer’s disease (AD) is a progressive neurodegenerative disorder characterized by amyloid-beta (Aβ) plaques, neurofibrillary tangles, neuroinflammation, and metabolic dysfunction, ultimately leading to cognitive decline and dementia. Emerging research highlights the microbiota-gut-brain axis as a crucial factor in AD pathogenesis, with gut dysbiosis contributing to neuroinflammation, immune dysregulation, and blood-brain barrier permeability. Microbial metabolites, such as short-chain fatty acids and amyloid-like proteins, can influence neurodegeneration, either exacerbating or mitigating disease progression. These findings position the microbiome as a key target for potential interventions, including dietary modifications, probiotics, and fecal microbiota transplantation (FMT), while salivary microbiome analysis presents a promising avenue for non-invasive AD diagnostics. Understanding and standardizing microbiome signatures may provide critical insights into disease mechanisms and open new pathways for early detection and therapeutic strategies.

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